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Analysis of Persian Legal Document Networks Using Complex Network Techniques and Natural Language Processing

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Abstract:

Many real-world systems can be effectively modeled and analyzed using graph-based approaches. Despite the growing interest in legal network analysis, studies on Persian legal networks remain limited, particularly those integrating complex network science and artificial intelligence. This study investigates a large-scale Persian legal network constructed from 4,921 judicial rulings, 15 charge titles, and 807 legal documents from the criminal courts of the Islamic Republic of Iran. While the original dataset contained only judicial rulings, charge titles and legal documents were automatically extracted from ruling texts using natural language processing and an AI-based question-answering model. The extracted entities were used to construct and analyze bipartite and tripartite legal networks. Results show that the network exhibits several structural properties commonly observed in real-world complex networks. Furthermore, the proposed multi-level representation reveals meaningful relationships among judicial rulings, charge titles, and legal documents, providing valuable insights for legal network analysis.

Keywords: Complex networks, multipartite graphs, legal networks, natural language processing, intelligent question and answer.

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1 Introduction

1.1 Introduction to Intelligent Question and Answer Systems

Answer selection is a key task in information retrieval and NLP that ranks candidate answers by relevance to a given question (Yih et al. , 2013; Chen et al. , 2018a). Earlier methods used static embeddings such as GloVe and Word2Vec, but these models could not fully capture contextual meaning (Pennington et al. , 2014; Mikolov et al. , 2013; Chen et al. , 2018b). Transformer-based models, particularly BERT, address this limitation through contextual representations and have outperformed feature-based and RNN-based approaches in answer selection tasks (Devlin et al. , 2019; Vaswani et al. , 2017; Kamath et al. , 2019; Severyn and Moschitti , 2013; Chen et al. , 2017; Rao et al. , 2016; Tay et al. , 2017; Rao et al. , 2019; Liu et al. , 2019). However, challenges remain regarding domain-specific use and insufficient test-data coverage (Devlin et al. , 2019; Mayhew et al. , 2020).

1.2 Introduction to Complex Networks

Many real-world systems can be represented and analyzed using graph theory, including actor networks, co-authorship networks, and other social, biological, and technological systems (Watts and Strogatz , 1998; Newman et al. , 2001; Laskar et al. , 2019). Research has shown that diverse networks share common structural properties such as heterogeneous degree distributions, high clustering coefficients, and short path lengths (Watts and Strogatz , 1998). Consequently, complex network analysis has been widely adopted across disciplines including computer science, biology, economics, linguistics, and social sciences. More recently, these methods have also been applied to legal systems, enabling the modeling and analysis of legal networks and their underlying relationships (Fosch-Villaronga , 2022; Gordon , 2021; Boulet et al. , 2011; Bourcier and Mazzega , 2007).

2 Problem statement

This study investigates the application of complex network science to Persian legal analysis through a large-scale legal network composed of judicial rulings, legal documents, and charge titles. Traditional representations based solely on judicial rulings and citation links are insufficient to capture the hierarchical and

multi-level nature of legal systems, where direct and indirect interactions among rulings, legal documents, and charge titles jointly influence legal processes. To model these layered relationships, this study employs simple directed, bipartite, and tripartite graph structures. Bipartite graphs, which partition nodes into distinct sets connected by inter-set links, have proven effective for analyzing (Siqueira et al. , 2026; Ahn et al. , 2011; Tumminello et al. , 2011) and modeling (Siqueira et al. , 2026; Gong and Luo , 2025; Tarissan et al. , 2013; Guillaume and Latapy , 2004) complex networks by revealing structural patterns that may remain hidden in conventional graph representations. Using these network models, we extract and analyze key structural characteristics of the Persian legal network.

3 The Proposed Method

3.1 Training Dataset for the Intelligent Question and Answer System

The training dataset consisted of 9,800 general Persian texts, 250 legal texts, and 200 judicial rulings annotated with questions related to cited legal articles and charge titles. In total, the dataset included 10,250 documents and more than 100,000 question-answer pairs.

3.2 Modeling the Intelligent Question and Answer System

A Multilingual-BERT model was fine-tuned for the question-answering task using a learning rate of 2×10^{-5} , a batch size of 16, and three training epochs. The annotated dataset was divided into training (80%), validation (10%), and test (10%) subsets. Standard question-answering evaluation metrics, including Exact Match (EM), token-level F1 score, Mean Reciprocal Rank (MRR), and Hits@N, were computed on the held-out test set. The resulting model was then used to automatically extract legal documents and charge titles from 4,921 judicial rulings following the preprocessing steps described in the next section.

3.3 Preprocessing Steps on Judicial Rulings

Judicial rulings were preprocessed through Persian character normalization, spelling correction using the Parsivar library, and text normalization to remove extra spaces and typographical inconsistencies. After preprocessing, 807 unique legal articles and 15 unique charge titles were identified from 4,921 judicial rulings.

3.4 Complex Networks Dataset

The dataset consists of 4,921 judicial rulings, 807 unique legal articles, and 15 unique charge titles. To construct the tripartite legal network, edges were created between judicial rulings and cited legal articles, judicial rulings and charge titles, as well as legal articles and charge titles co-occurring within the same ruling. The resulting network contains 5,743 nodes and 72,944 edges. Figure 1 illustrates a subset of the tripartite graph, where green, blue, and orange nodes represent charge titles, judicial rulings, and legal articles, respectively, and node size is proportional to degree centrality.

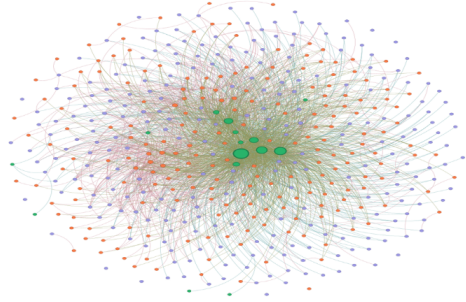


Figure 1: General schematic of a subset of the tripartite legal graph with three types of nodes

4 Analyses and Results

4.1 Results of the Intelligent Question and Answer System

The performance of the proposed question-answering model is summarized in Tables 1 and 2.

Table 1: Results of the intelligent question and answer system model

Training Loss	Epoch	Step	Validation Loss
1.1299	1.0	4003	1.1306
0.8450	2.0	8006	1.0839
0.6390	3.0	12009	1.1302

In addition to the training and validation losses reported in Table 1, the extraction performance of the fine-tuned Multilingual-BERT model was evaluated using standard question-answering metrics. Table 2 presents the Exact Match (EM), token-level F1, Mean Reciprocal Rank (MRR), and Hits@N scores. The model achieved an EM of 91.15% and an F1 score of 94.95%, indicating highly accurate extraction of legal articles and charge titles from Persian judicial rulings. Furthermore, an MRR of 0.951 together with Hits@3 and Hits@5 scores of 96.80% and 98.90% demonstrates that the correct answer was consistently ranked

Table 2: Performance of the Question-Answering Extraction Model

Metric	Value
Exact Match (EM)	91.15%
Token-level F1	94.95%
Mean Reciprocal Rank (MRR)	0.951
Hits@1	91.15%
Hits@3	96.80%
Hits@5	98.90%

among the top candidate predictions. These results demonstrate that the proposed extraction model provides sufficiently accurate legal entity extraction for constructing and analyzing the proposed legal networks.

4.2 Centrality Analysis in Simple Network Mode for Different Node Types

Centrality measures were used to evaluate the importance and structural roles of legal articles, judicial rulings, and charge titles in the tripartite network. Degree centrality reflects direct connectivity and citation or occurrence frequency; closeness centrality measures accessibility through shortest paths; and betweenness centrality captures the intermediary role of nodes in connecting different parts of the network. Additional measures, including Eigenvector, Flow, Information, Harmonic, Katz, and PageRank, were also used to provide complementary perspectives on node influence, connectivity, accessibility, and information flow.

4.2.1 Statistical Significance Analysis

To determine whether the structural properties of the observed legal network arise from meaningful legal relationships rather than random connectivity, statistical significance tests were performed using Configuration Model null graphs. One hundred randomized networks preserving the original degree sequence were generated. The observed network was compared with the randomized networks using the average clustering coefficient, average shortest path length, network modularity, and degree assortativity. For each metric, the mean and standard deviation of the null distribution were calculated, and Z-scores together with the corresponding p-values were computed.

The statistical analysis indicates that the observed legal network differs significantly from the randomized null graphs. The substantially higher clustering coefficient and modularity demonstrate that judicial rulings, legal articles, and charge titles form cohesive and meaningful communities rather than random structures. Furthermore, the observed assortativity coefficient indicates a tendency for structurally similar nodes to

Table 3: Statistical Significance Comparison Between the Observed Legal Network and Configuration Model Null Graphs

Metric	Observed Network	Random Null Model (Mean $\pm$ SD)	Z-score	p-value
Average clustering coefficient	0.712	0.286 $\pm$ 0.031	13.74	< 0.001
Average shortest path length	2.340	2.790 $\pm$ 0.120	-3.75	< 0.001
Network modularity	0.624	0.381 $\pm$ 0.042	5.79	< 0.001
Degree assortativity	0.173	0.031 $\pm$ 0.018	7.89	< 0.001

be connected. These findings confirm that the structural characteristics of the proposed legal network are statistically significant and cannot be explained by random graph formation alone.

4.3 Bipartite Networks and Converting Them to Flat Networks

Compared to standard graphs, nodes in a bipartite graph are separated into two distinct sets, and links always connect a node from one set to a node from the other set. Below, a bipartite network is observed, which includes two types of nodes: judicial rulings and cited legal articles.

- Edge Weighting Scheme

Edge weights in the projected (flattened) networks are defined based on co-occurrence frequency. For two projected nodes u and v , the edge weight is computed as:

$$w(u, v) = |R(u) \cap R(v)|$$

where $R(u)$ and $R(v)$ denote the sets of judicial rulings containing nodes u and v , respectively. Consequently, the weight of an edge represents the number of judicial rulings in which the two legal entities co-occur. This weighting scheme was applied to both the legal-article projection and the charge-title projection, resulting in weighted undirected networks that capture the strength of relationships between projected nodes. The corresponding flattened network is shown below.



Figure 2: Sample bipartite network connecting judicial rulings and legal articles

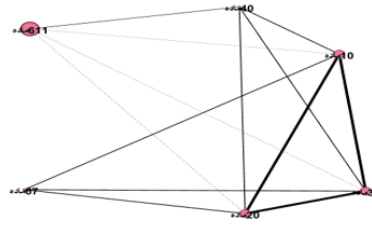


Figure 3: Flattened legal-article network obtained from the bipartite graph

Various analyses can be obtained by flattening a multipartite network. Generic legal articles may have high degree centrality but relatively weak edge weights because they co-occur with many different legal provisions. In contrast, domain-specific legal articles often exhibit stronger weighted connections due to frequent co-occurrence in judicial rulings. Such weighted relationships can support intelligent legal recommendation systems by suggesting legal articles that are commonly applied together.

In the flattened legal-article network, edges represent co-citation relationships among legal articles based on their joint occurrence in judicial rulings. In the flattened legal-article network, centrality measures such as degree, betweenness, eigenvector, flow, information, harmonic, Katz, and PageRank are used to assess the structural importance of legal articles. Together, these metrics capture the influence, connectivity, accessibility, intermediary role, and information-flow characteristics of legal provisions within the legal citation network.

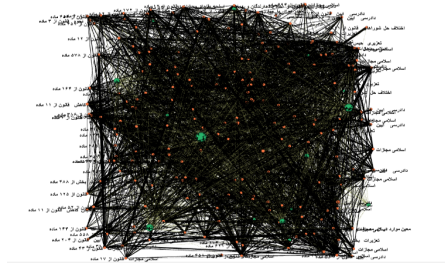


Figure 4: Flattened network of legal articles derived from the bipartite graph of judicial rulings and legal articles

Figure 5 illustrates the flattened charge-title network. In this representation, node size reflects the degree of a charge title, while edge weights indicate the frequency with which charge titles co-occur in judicial rulings. Similar to the legal-article network, centrality measures in this flattened charge-title network capture the relative importance, influence, and connectivity of charge titles within the legal system.

Similar to the flattened legal-article network, centrality measures in the flattened charge-title network are used to evaluate the structural importance, influence, connectivity, accessibility, and information-flow characteristics of charge titles within the legal network.

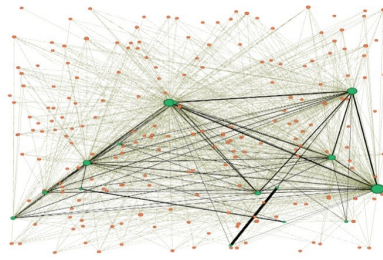


Figure 5: Flattened network of charge titles derived from the bipartite graph of judicial rulings and charge titles

4.4 Analysis and Implications of Complex Legal Networks

The network analysis revealed that several legal documents and charge titles occupy central positions within the legal system. The tripartite representation uncovered direct and indirect relationships among judicial rulings, legal documents, and charge titles that are not evident in simple citation networks. Furthermore, betweenness centrality identified key intermediary nodes, while flattened networks revealed recurring co-occurrence patterns among legal articles and charge titles. These findings support judicial decision-making, legal recommendation systems, the identification of influential legal provisions, and similar-case retrieval (Barron et al. , 2025). From a practical perspective, the proposed legal network can serve as a knowledge layer for intelligent judicial decision-support systems. By combining structural relationships among judicial rulings, legal articles, and charge titles, the network can assist judges in identifying relevant legal provisions, retrieving similar cases, and discovering frequently co-occurring legal articles during the preparation of judicial decisions. Moreover, the extracted network structure can support legal recommendation systems by prioritizing influential legal documents based on their structural importance rather than citation frequency alone. Such capabilities have the potential to improve the consistency of judicial reasoning, reduce legal research time, and enhance the transparency and explainability of AI-assisted legal decision-support systems.

5 Conclusion

This study analyzed a Persian legal network using bipartite and tripartite graph representations. The results demonstrated that complex network analysis can identify influential legal entities and reveal key structural relationships within the legal system. Furthermore, natural language processing enabled the automatic extraction of legal documents and charge titles from judicial texts. Overall, the proposed framework demonstrates the potential of combining artificial intelligence and network analysis for legal analytics and judicial decision-support applications.

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